Assignment No. 16

Analysis on Mixing Elbow

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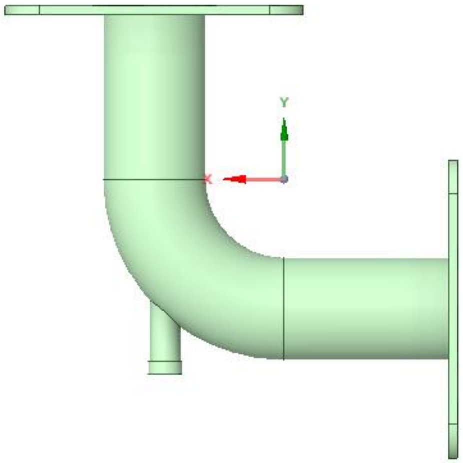
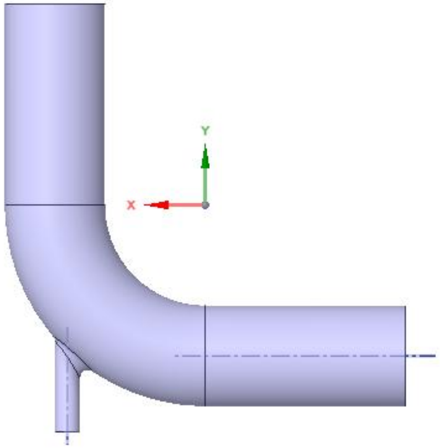
Objective

- Objective of this study is to analyze the quality of mixing of a mixing elbow. The study aims to acquire the temperature distribution on midplane and outlet of the mixing elbow, velocity vector profile on the midplane, mass flow rate, average outlet temperature and velocity.

Geometry details

- A computer aided design was already provided, from which volume was extracted. The diameter of the larger tube is approximately 10.16 cm and the tube spans for about 43.254 cm (from inlet/outlet to the curve)



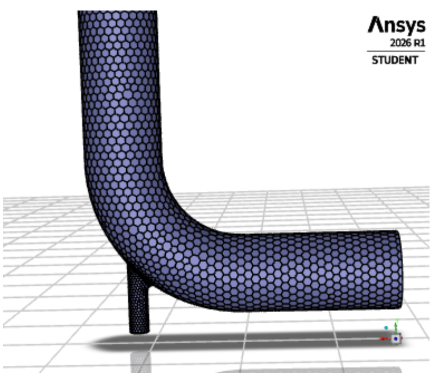


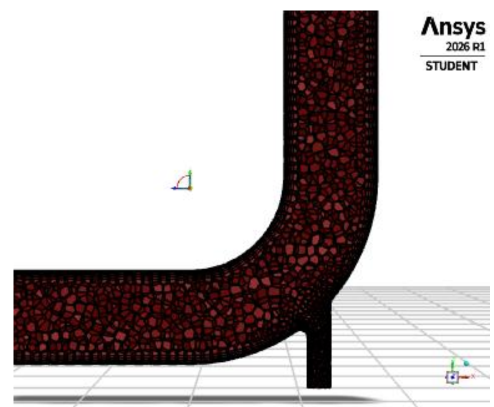
Boundary Conditions

- Velocity and temperature for larger inlet is respectively 20 m/s and 60 degree C. Velocity and temperature for smaller inlet is respectively 30 m/s and 100 degree C. The oulet is kept is ambient pressure and the gravity effect is ignored .

Blocking Details

- The extracted volume was blocked in ANSYS Fluent mesher using ‘watertight’ workflow. 4 boundary layers, with growth rate of 1.2 , were added to ensure smooth transition. Volume mesh was prepared using polyhedra. 2 cores were used in the process of a 4 core CPU Intel core i5(3.3 GHz) , 4 GB ddr3 desktop.
- Cells: 17670, Nodes: 58625





CFD settings

- ANSYS Fluent 2026 R1 student version was used for performing the simulation
- 3 cores were used for parallel processing, thus number of partitions was 3.
- The Mesh Quality was excellent with orthogonal quality of 6.27e-01.
- Solver: Pressured Based, no gravity, absolute velocity, time steady
- Energy Equation was turned on as temperature was involved
- Viscous: K-omega(2 eqn) SST model was used

- Material: Water
- BC: As discussed in the geometry details, keeping the outlet at ambient pressure(gauge pressure zero) and the backflow temperature at 60 degree C(only for numeric stability ,no physical meaning). Wall was stationary, no slip and with zero heat flux(adiabatic)
- Solution: Pressure Velocity Coupling: SIMPLE

Gradient: Least Square Cell based

Pressure: 2^{nd} order

Momentum, Turbulent KE, Specific Dissipation Rate, Energy: 2^{nd} order upwind

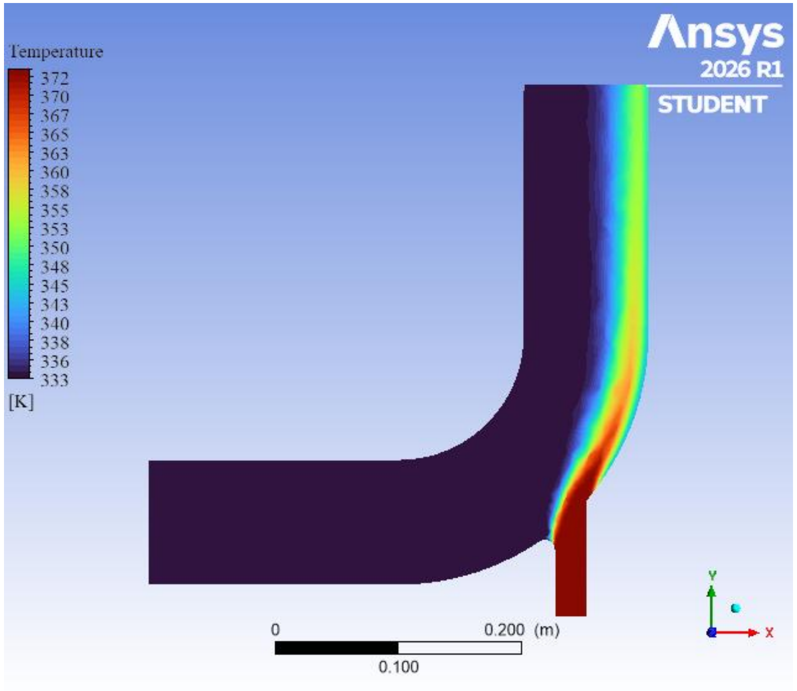
Initialization and Calculation

- Initialization: Hybrid Initialization
- Calculation: 200 iterations were offered, solution converged at 68

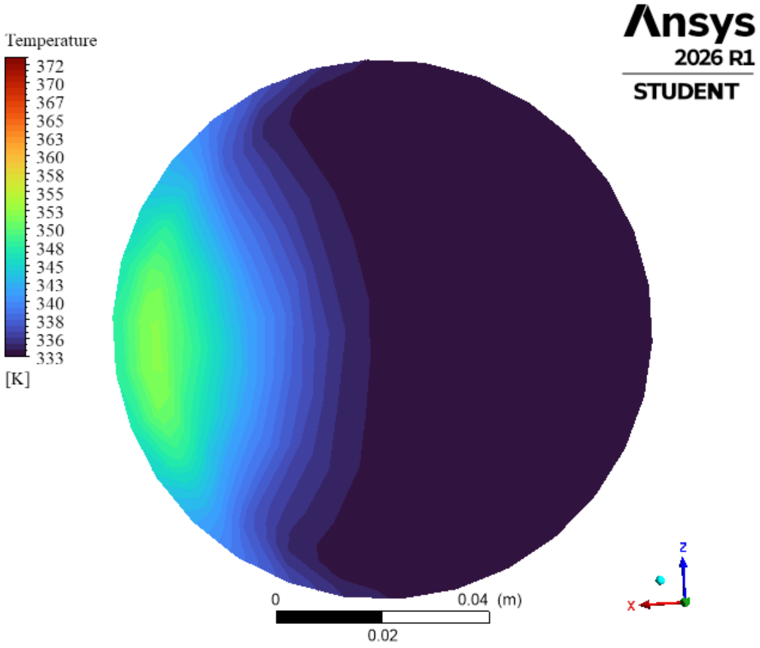
Results

For post processing CFD post of CFX was used

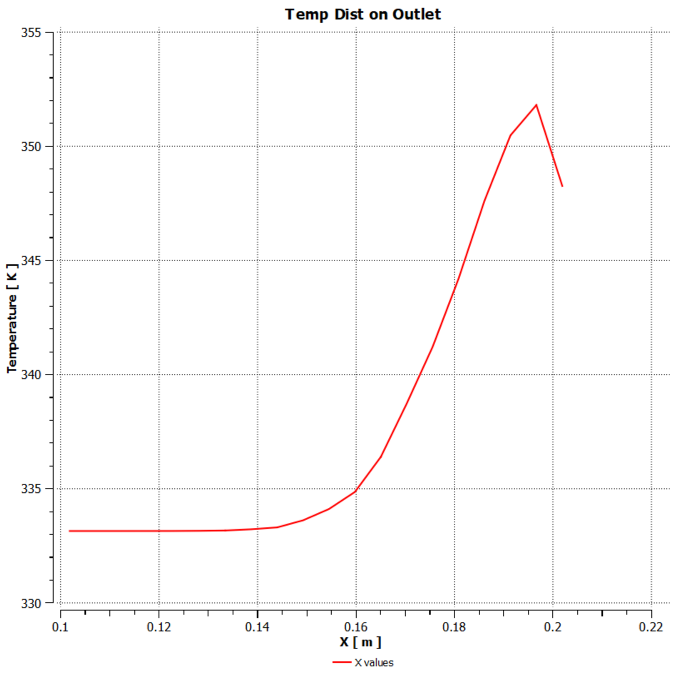
Temperature Contour at midplane



Temperature contour at outlet



Temperature distribution graph at outlet(midplane):



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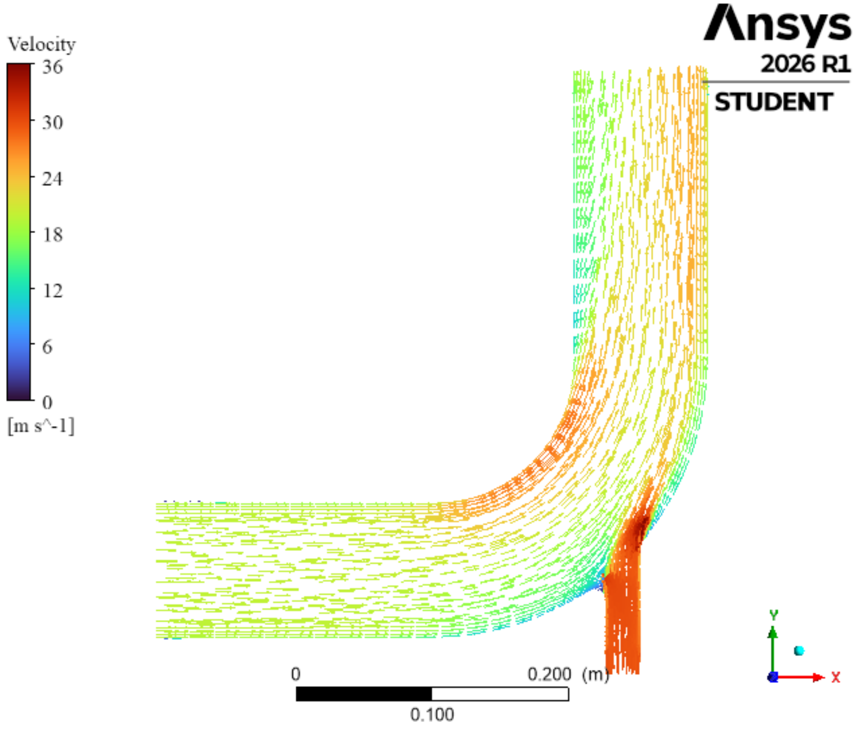
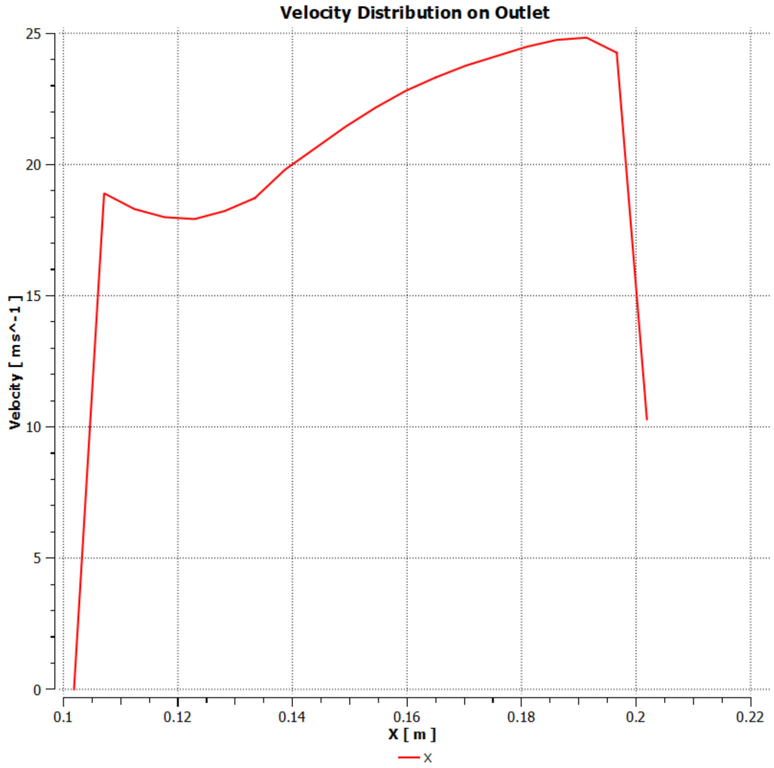
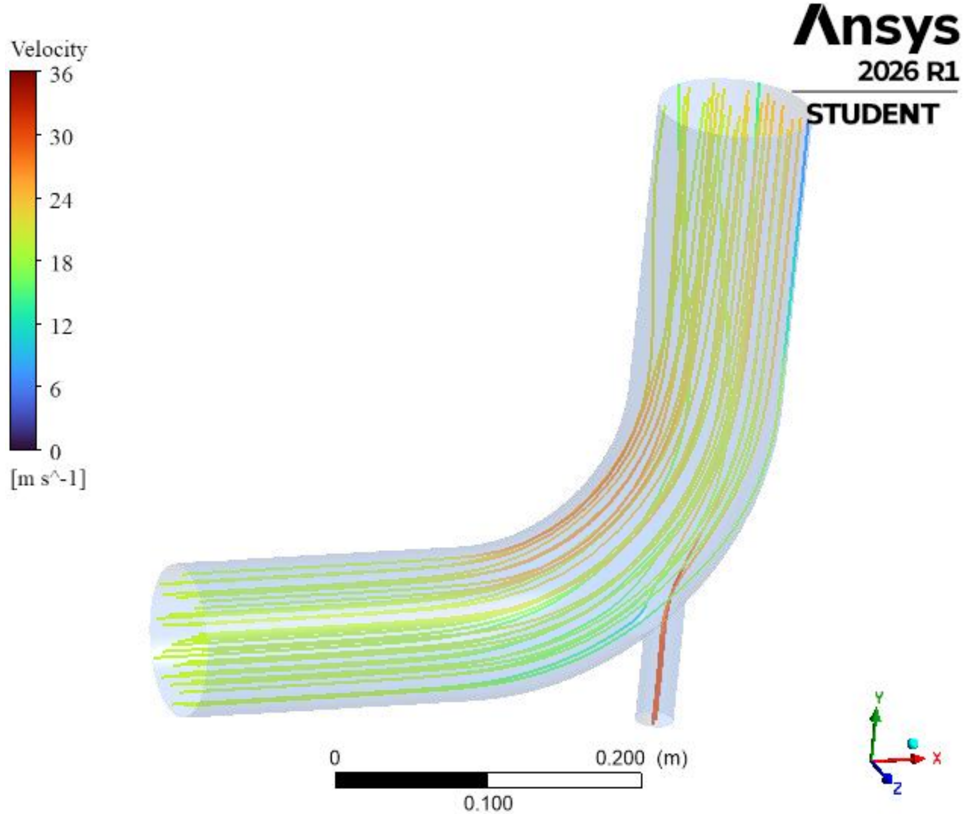
Velocity vectors at midplane



Velocity Distribution graph at outlet(midplane)



3D streamline inside mixing elbow



- Mass flow rate = 173.477 kg/s
- Avg outlet velocity = 21.8755 m/s
- Avg outlet temperature = 336.223 K

WorkBench was not used. All of the softwares were run as stand alone

Conclusions

- From the provided graphs and contours , it can be said that the fluids of different temperature and velocity is not properly mixed. The outlet velocity and temperature distribution graph shows, the mixing is far from optimal. Twisted metal pieces can be placed in the pipe for better mixing.